

AI Research Assistant - Architecture Design

1. Introduction

The AI Research Assistant is designed using a modular, agent-based architecture to effectively perform academic and scientific research tasks. The system is highly decoupled to allow easy incorporation of new tools, language models, and external databases.

2. Core Components

Agent Orchestrator: The central coordinator that manages the lifecycle of research tasks, sub-agents, and task delegation.

Memory Subsystem: Responsible for context retention across long-running tasks. Integrates short-term working memory and long-term vector-based retrieval (RAG).

Tool Registry: A plugin-based system where various external utilities (web search, document parsers, code interpreters) are registered.

Document Pipeline: Ingests and parses multiple file formats (PDF, HTML, DOCX), extracting structured data, tables, and text.

Report Generator: Synthesizes gathered knowledge into formatted markdown or PDF documents, automatically managing citations.

3. Typical Workflow

1. User provides a research query or hypothesis.
2. Planner Agent breaks down the query into achievable steps.
3. Search & Retrieval Agent queries external knowledge bases and academic databases.
4. Analysis Agent reviews retrieved data, extracts insights, and updates memory.
5. Report Agent synthesizes the findings into a comprehensive research artifact.

4. Deployment Architecture

The system is fully containerized using Docker, allowing it to run seamlessly on local workstations or in scalable cloud environments. Services are orchestrated via docker-compose.